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Huang, Kaixing, Zhou, Chunjie, Tian, Glen, Tu, Weixun, & Peng, Yuan (2017)
Application of Bayesian network to data-driven cyber-security risk assessment in SCADA networks.
In Gregory, M A (Ed.) *Proceedings of the 2017 27th International Telecommunication Networks and Applications Conference (ITNAC).*
Institute of Electrical and Electronics Engineers Inc., United States of America, pp. 96-101.

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https://doi.org/10.1109/ATNAC.2017.8215355

Application of Bayesian Network to Data-Driven Cyber-Security Risk Assessment in SCADA Networks

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***Abstract*—Supervisory control and data acquisition (SCADA) systems are the key driver for critical infrastructures and industrial facilities. Cyber-attacks to SCADA networks may cause equipment damage or even fatalities. Identifying risks in SCADA networks is critical to ensuring the normal operation of these industrial systems. In this paper we propose a Bayesian network-based cyber-security risk assessment model to dynamically and quantitatively assess the security risk level in SCADA networks. The major distinction of our work is that the proposed risk assessment method can learn model parameters from historical data and then improve assessment accuracy by incrementally learning from online observations. Furthermore, our method is able to assess the risk caused by unknown attacks. The simulation results demonstrate that the proposed approach is effective for SCADA security risk assessment.**

***Index Terms*—cyber-security; risk assessment; SCADA network; Bayesian network; machine learning.**

I. Introduction

Supervisory control and data acquisition (SCADA) systems are used to monitor and control critical processes for industrial applications, such as smart Grids, waste water treatment, gas delivery, etc. SCADA networks are highly vulnerable to cyber-attacks as they are adopting more and more information and communication technologies [1]. Attacks against SCADA networks could cause not only information leakage but also physical destruction or even fatalities. Therefore, the security of SCADA networks is an issue of great concern.

Cyber-security risk assessment (CSRA) is the process of identifying the likelihood of cyber-attack and the impact it would have. CSRA plays a vital role in the security protection of SCADA networks since system managers need to take intrusion defense actions based on the risk assessment results. Generally there are two primary types of CSRA: qualitative and quantitative. Quantitative methods are preferred here because they can provide more accurate reflection of system status, thus facilitating better allocation of safeguard resources. On the other hand, CSRA can also be categorized into another two types: offline and online. Offline CSRA is to create an inventory for all system assets, find the vulnerabilities of components, predict attack scenarios and estimate the difficulty for attackers to success. Online CSRA aims to evaluate real-time system risk level based on the alerts from intrusion detection systems. Online CSRA can provide immediate feedback to operators before the attacker has caused severe damages. Offline CSRA has been comprehensively studied in the literature while online CSRA has not gained enough attention, so we consider the latter one here.

Currently, online quantitative CSRA methods for SCADA networks are mostly based on the experience of security experts. However, since expert knowledge is usually incomplete and subjective, the assessment results might be inaccurate and these methods can not assess the risk caused by unknown attacks. Hence, there is a need for a novel CSRA approach to more accurately assessing the security risk of SCADA networks under known or unknown cyber-attacks.

In this paper we employ Bayesian networks (BN) to build a dynamic and quantitative CSRA model for SCADA networks. First, we integrate traditional BN-based security risk assessment methods with Leaky Noisy-OR (LNOR) gate to support risk assessment for unknown attacks. Then we use machine learning techniques to learn the model parameters from data in order to generate more accurate risk value. Two parameter learning algorithms are proposed for BNs extended with LNOR gate, and they are used for offline batch learning from historical data and online incremental learning from real-time attack evidences, respectively. Finally, a case study on a sample SCADA system is analyzed to illustrate the proposed approach.

The rest of this paper is organized as follows. Section II discusses some related work. Section III describes the BN-based CSRA model. Section IV presents the parameter learning algorithms for BNs. Section V shows the experiment results. Finally, we conclude in Section VI.

II. Related Work

BN is a useful probabilistic tool for risk assessment, which has been widely used in computer networks [2]. While there have been many CSRA methods proposed for SCADA networks, the applications of BNs for SCADA security risk assessment are relatively few.

Shin, Son and Heo [3] adopted BNs to evaluate security risk for nuclear SCADA systems. The analytical results from the BN model were fed into an event tree for safety assessment. Reference [4], [5] applied BNs to security risk assessment of power grid SCADA networks. Their approach could predict the probabilities of successful attacks and analyze the reliability of power systems. Zhang et al. [6] proposed a multi-model-based CSRA method for industrial SCADA systems. In this study, attack tree, fault tree and event tree were integrated to construct a BN model for unified risk assessment.

Though the above studies have provided some initial work to apply BNs to CSRA in SCADA networks, all of them built the assessment models with expert experience. However, expert knowledge is usually incomplete and subjective, and systems may evolve dynamically with time, so the ability of self-learning is necessary as it can improve the assessment accuracy and help CSRA models adapt to system changes.

III. Proposed Risk Assessment Method

In this section, the architecture of SCADA networks is introduced first, and then the proposed BN-based CSRA model is presented. Finally, we describe how the security risk assessment process is carried out using the proposed model.

A. Architecture of SCADA networks

SCADA systems are widely used in industrial sectors to monitor and control distributed components. A typical SCADA network consists of three layers: corporate layer, supervisory layer and field control layer, as shown in Fig. 1. SCADA networks have many kinds of devices, such as PC, historical database, human machine interface (HMI), engineering station, master terminal unit (MTU), remote terminal unit (RTU). HMI, engineering station and database are usually located in a control center for unified management. MTUs are responsible for distributing control commands to the field control layer and transferring sensor data to the control center. RTUs are remote field control devices which interact with physical processes. As SCADA networks are adopting more and more open information technologies, vulnerabilities are inevitable in these systems.

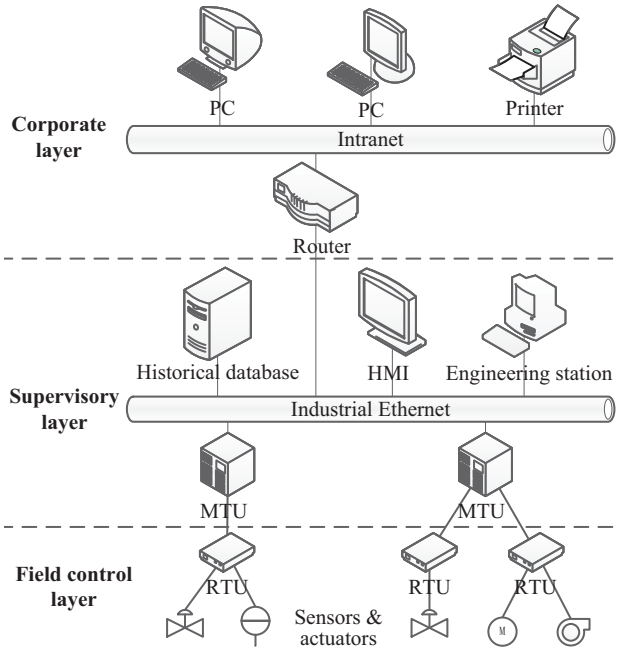


Fig. 1. Architecture of SCADA networks.

B. BN-based risk assessment model

A BN is a directed acyclic graph where the nodes represent variables and the directed arcs imply the conditional dependencies between these variables. Each node in a BN is only dependent on its parent nodes, which is described by a conditional probability table (CPT). Given a BN $\mathbf{B}$ which has a set of variables $\mathbf{X} = \{X_1, X_2, ..., X_N\}$, the joint probability distribution of $\mathbf{X}$ is

$$P(\mathbf{X}) = P(X_1, X_2, ..., X_N) = \prod_{i=1}^{N} P(X_i|pa(X_i)), \quad (1)$$

where $pa(X_i)$ is the parent set of variable X_i.

When BNs are applied to CSRA, the network structure must be constructed first. Since attackers generally need to exploit system vulnerabilities to conduct various kinds of attacks for privilege escalation, we use two kinds of nodes to build the BN structure and they are vulnerability exploited (VE) node and administrator privilege gained (APG) node. Each APG node is associated with a host, and the VE nodes represent vulnerability information. By performing vulnerability scanning or querying the CVE database [7], the BN structure can be constructed. A sample BN model is illustrated in Fig. 2.

After the BN structure has been built, the CPT parameters need to be estimated. Though experts can guess the parameters directly, we suggest that learning from data could be a better choice because it can generate more accurate risk

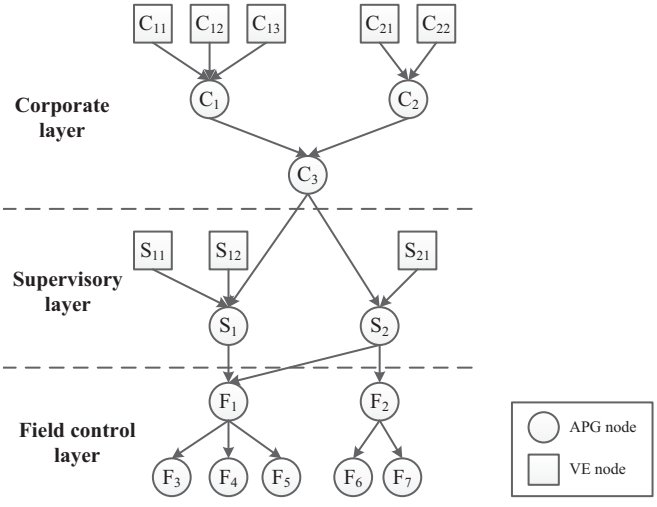


Fig. 2. BN-based CSRA model for SCADA networks.

values. The parameter learning issues are discussed in section IV.

C. Risk assessment using BNs

According to Kaplan et al. [8], risk can be quantified by

$$R = \sum_{i=1}^{M} V_{A_i} \times P(A_i), \tag{2}$$

where $\mathbf{A} = \{A_1, A_2, ..., A_M\}$ is the assets of SCADA networks, V_{A_i} is the value of A_i which can be quantified by the monetary value of A_i or its impact on the physical processes, and $P(A_i)$ is the probability of A_i being lost. In SCADA networks, the field control devices which are directly interacting with physical world are much more important than others because they could lead to disastrous consequences once compromised. Therefore, we only calculate the risk values for these devices. Consequently, each A_i corresponds to a leaf node in the field control layer of the proposed BN.

In order to conduct dynamic risk assessment, evidences need to be collected from intrusion detection systems in real time. Given a set of evidences $\mathbf{E} = \{E_1, E_2, ..., E_K\} \subset \mathbf{X}$, each E_i indicates that the corresponding ith BN node has been compromised by the attacker. Thus, we need to determine the probability of success for $\mathbf{A}$. The Bayes theorem is used to update the posterior probability:

$$P(A_i|\mathbf{E}) = \frac{P(\mathbf{E}|A_i) \times P(A_i)}{P(\mathbf{E})}, \tag{3}$$

where $P(A_i|\mathbf{E})$ is the conditional probability of A_i given $\mathbf{E}$, and $P(\mathbf{E}|A_i)$ is the conditional probability of $\mathbf{E}$ given A_i. Hence, the real-time risk value given evidence $\mathbf{E}$ can be calculated by:

$$R = \sum_{i=1}^{M} V_{A_i} \times P(A_i|\mathbf{E}). \tag{4}$$

IV. Parameter Learning

In this section, we present the offline batch learning and online incremental learning algorithms.

A. Leaky Noisy-OR gate

Tractability of CPT construction is generally referred to as the bottleneck of BN modeling due to the large number of CPT parameters and the absence of enough sample data. The primary solution to this problem is the canonical gate, which can significantly reduce the number of CPT parameters by taking advantage of independence of causal influence (ICI). Here we adopt the Leaky Noisy-OR gate to alleviate the difficulty of learning BN parameters. LNOR gate assumes that all the causes of an effect can produce the effect independent of others and there also exists the possibility that the effect can be produced even if all the causes are absent. When integrated with BN-based CSRA model, LNOR gate is able to describe the situations in which the BN model does not capture all the vulnerabilities in the system. In other words, with LNOR gate, the BN model can assess the risk caused by unknown attacks which exploits unidentified vulnerabilities.

Let there be Q binary variables $\mathbf{X}_q = \{X_1, X_2, ..., X_Q\}$ and they are the causes of effect Y. The LNOR gate can be modeled by adding an additional variable X_L, called the leaky variable, into $\mathbf{X}_q$. Let p_i be the probability that Y is true when X_i is true and all other causes are false. Then the probability of $Y = 1$ in the leaky model is given by

$$P(Y = 1|\mathbf{X}_q) = 1 - (1 - p_L) \prod_{i=1}^{Q} (1 - p_i), \tag{5}$$

where p_L is the probability that Y is true when X_L is true and all other variables are false.

B. Offline batch learning

Suppose that we already have a complete dataset offline, then the maximum likelihood estimate or Bayesian estimate method can be directly utilized for learning the BN parameters [9]. However, it is quite common that the attack sample dataset is incomplete in SCADA systems. This is because some variables defined in the BN model might be unobservable when the compromised hosts do not respond to queries. To address this data incomplete problem, the expectation maximization (EM) algorithm is the most common solution. The principle of EM algorithm is to fill the missing values in the incomplete dataset using the inference algorithms of BN. In this study, we have proposed a BN model with binary variables and extended it with LNOR, so some modifications are needed to adapt the EM algorithm to our case.

Given a BN $\mathbf{B} = (\mathbf{X}, \Theta)$ where $\mathbf{X} = \{X_1, X_2, ..., X_N\}$ represents a set of binary variables and Θ represents the CPT parameters, we first initialize Θ with random value Θ^0, and then generate new Θ iteratively. Let θ_{ih} be the probability of $X_i = 1$ when $pa_h(X_i) = 1$ and $\forall j \neq h, pa_j(X_i) = 0$, where $pa_h(X_i)$ is the hth parent node of X_i. Suppose that Θ^s is the CPT parameters obtained at the sth iteration, then the $(s+1)$th iteration includes four steps:

1) *Data completion:* Fill the missing values using (5).
2) *Calculation of expectation:* Calculate the maximum likelihood estimate of Θ^{s+1} using the repaired dataset via

$$\theta_{ih}^{s+1} = \begin{cases} \frac{m_{ih}}{\sum_{k=1}^{r_i} m_{ihk}}, & if \ \sum_{k=1}^{r_i} m_{ihk} > 0 \\ \frac{1}{r_i}, & otherwise \end{cases} \tag{6}$$

   where m_{ih} is the number of case "$X_i = 1, pa_h(X_i) = 1; \forall j \neq h, pa_j(X_i) = 0$" in the repaired dataset, r_i is the number of possible values of X_i, and m_{ihk} is the number of case "$X_i = k, pa_h(X_i) = 1; \forall j \neq h, pa_j(X_i) = 0$".
3) *Repeat:* Repeat step 1-2 until the accuracy is reached.
4) *Calculation of CPT parameters:* In the leaky BN model, the CPT parameters estimated using (6) also include the influence of the leaky node [10], i.e.:

$$\theta_{ih} = 1 - (1 - \theta_{iL})(1 - \theta_{ih}^{'}), \tag{7}$$

   where $\theta_{ih}^{'}$ is the probability of $X_i = 1$ when $pa_h(X_i) = 1, pa_L(X_i) = 0$ and $\forall j \neq h, pa_j(X_i) = 0$, and θ_{iL} is

the leaky probability which can be estimated from the dataset as the proportion of the cases in which $X_i = 1$ among those in which all the parent nodes of X_i is 0. Then we can get

$$\theta'_{ih} = 1 - \frac{1 - \theta_{ih}}{1 - \theta_{iL}}, \tag{8}$$

Finally, the probability of $X_i = 1$ is given by

$$P(X_i = 1) = 1 - (1 - \theta_{iL}) \prod_{i=1}^{H} \frac{1 - \theta_{ih}}{1 - \theta_{iL}}, \tag{9}$$

where H is the number of parent nodes of X_i.

C. Online incremental learning

Online incremental learning of BN parameters is to update the model by processing observations one at the time as they arrive. The Voting EM algorithm [11] is adapted here to online parameter learning of BNs extended with LNOR gate.

Suppose that there has been an available dataset $\mathbf{Z} = \{Z_1, Z_2, ..., Z_{t-1}\}$ and Z_t is a new observed record at time t. Let Θ^{t-1} denote the parameters of a BN $\mathbf{B} = (\mathbf{X}, \Theta)$ at time $(t-1)$. Reference [12] pointed out that the updating of Θ at time t can be realized by the following maximization:

$$\Theta^t = argmax_\Theta[\eta Lo(\Theta|\mathbf{Z}, Z_t) - d(\Theta, \Theta^{t-1})], \tag{10}$$

where $Lo(\cdot)$ denotes the log likelihood, η is the learning rate and $d(\Theta, \Theta^{t-1})$ calculates the Chi squared distance between Θ and Θ^{t-1}. By solving (10) with the method proposed in [12], we can get the updating rule:

$$\theta^t_{ih} = \begin{cases} \theta^{t-1}_{ih} + \eta[\frac{P(pa_h(X_i)=1, X_i=1|Z_t, \Theta^{t-1})}{P(pa_h(X_i)=1|Z_t, \Theta^{t-1})} - \theta^{t-1}_{ih}], \\ \quad if\ P(pa_h(X_i) = 1|Z_t, \Theta^{t-1}) > 0 \\ \theta^{t-1}_{ih}, \quad otherwise \end{cases} \tag{11}$$

where θ^t_{ih} is the parameter θ_{ih} at time t. When Z_t is a complete observation, (11) reduces to:

$$\theta^t_{ih} = \begin{cases} \eta + (1-\eta)\theta^{t-1}_{ih}, \ if\ X^t_i = 1, pa^t_h(X_i) = 1 \\ \quad and\ \forall j \neq h, pa^t_j(X_i) = 0 \\ (1-\eta)\theta^{t-1}_{ih}, \quad if\ X^t_i = 0, pa^t_h(X_i) = 1 \\ \quad and\ \forall j \neq h, pa^t_j(X_i) = 0 \\ \theta^t_{ih}, \quad otherwise \end{cases} \tag{12}$$

where X^t_i is the value of variable X_i in Z_t and $pa^t_h(X_i)$ is the value of $pa_h(X_i)$ in Z_t.

V. Case Study

In this section, we illustrate the proposed cyber-security risk assessment approach on a sample SCADA network.

A. A sample SCADA network

A chemical process SCADA network described in [6] is investigated here to illustrate the proposed approach. Fig. 3 shows the structure of the SCADA network which controls a chemical reaction process in a reactor. The SCADA network is comprised of a Firewall, a PC, a router (RO), a historical database (HD), an engineering station (ES), an HMI, two programmable logic controller (PLC1 and PLC2), four valves (V1, V2, V3 and V4), a temperature sensor (TS), a water level sensor (LS), a pressure sensor (PS), a switch (SW) and a motor (MO). In this scenario, PC has access to Internet through Firewall and it also can access HD, HMI and ES through RO; HD, HMI and ES communicate with PLC1 and PLC2 via Industrial Ethernet; PLC1 controls V1, V2, TS, LS and PS; and PLC2 controls V3, V4, SW and MO. V1 and V2 are feed valves; V3 and V4 are outlet valves; MO drives a blender; and SW controls a heater.

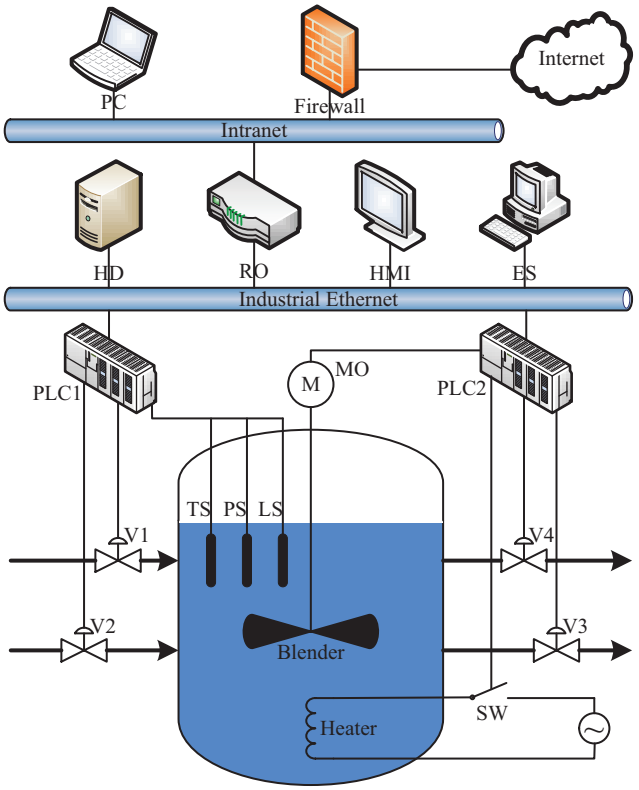


Fig. 3. A chemical process SCADA network.

B. Construction of the Bayesian network

The SCADA network shown in Fig. 3 has some vulnerabilities, as described in Table I. The PC has three remote procedure call (RPC) vulnerabilities (CVE-2008-4250, CVE-2013-5056 and CVE-2013-3175) which allow remote attackers to execute arbitrary code via a crafted RPC request. The HD has a database vulnerability (CVE-2016-7254) which allows remote attacks to gain privileges via unknown vectors. The ES has two software vulnerabilities in Siemens SCADA development kits (CVE-2013-0675 and CVE-2012-3015) which allow attackers to gain privileges. The PLC1 and PLC2 have the same vulnerability (CVE-2013-0659) which allows attackers to execute arbitrary code via UDP port 17185.

According to Table I, we can construct the Bayesian model, as shown in Fig. 4. In this figure, a circle represents the APG node for a host and a rectangle represents a vulnerability of a host. As discussed in section III-C, we need to know the value

TABLE I
VULNERABILITY INFORMATION

No.	Host	Vulnerability	Description
V_1	PC	CVE-2008-4250	Server Service Vulnerability
V_2	PC	CVE-2013-5056	Microsoft Scripting Runtime Object Library Vulnerability
V_3	PC	CVE-2013-3175	Remote Procedure Call Vulnerability
V_4	ES	CVE-2013-0675	Buffer Overflow in Siemens WinCC Server
V_5	ES	CVE-2012-3015	Untrusted Search Path Vulnerability in Siemens SIMATIC STEP7
V_6	HD	CVE-2016-7254	SQL RDBMS Engine Elevation of Privilege Vulnerability
V_7	PLC1	CVE-2013-0659	Arbitrary Code Execution on Siemens CP 1604
V_8	PLC2	CVE-2013-0659	Arbitrary Code Execution on Siemens CP 1604

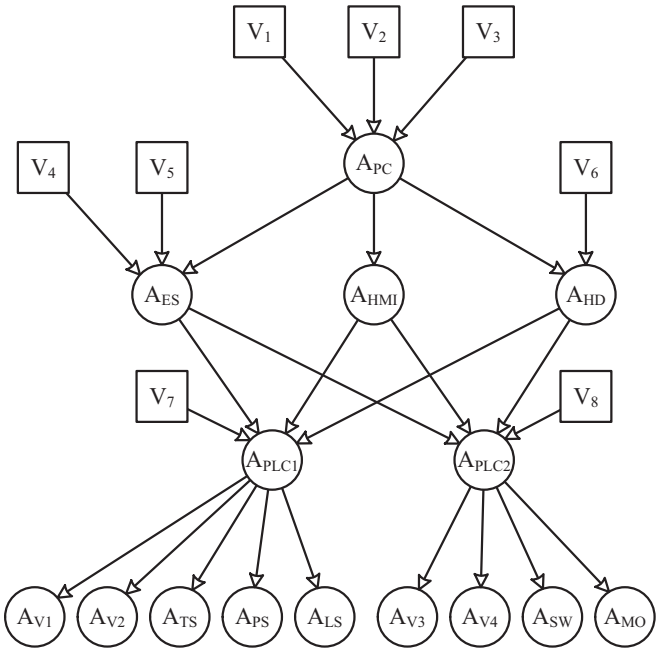


Fig. 4. Bayesian network for the chemical process SCADA network.

of the leaf nodes in the BN for quantitative risk assessment. For simplicity, we assume that they all have the same value of 10,000 dollars.

C. Simulation and result analysis

Three experiments are conducted in Matlab to verify the effectiveness of the proposed CSRA method, analyze the performance of the proposed learning algorithms, and evaluate the ability of risk assessment for unknown attacks, respectively.

The first experiment is conducted to verify the effectiveness of the proposed CSRA approach. Fig. 5 shows the risk curve of the SCADA network under a sample attack scenario. In this scenario, the attacker followed the five steps below:

1. Made use of Vulnerability V_1 to attack PC from the 10th minute.

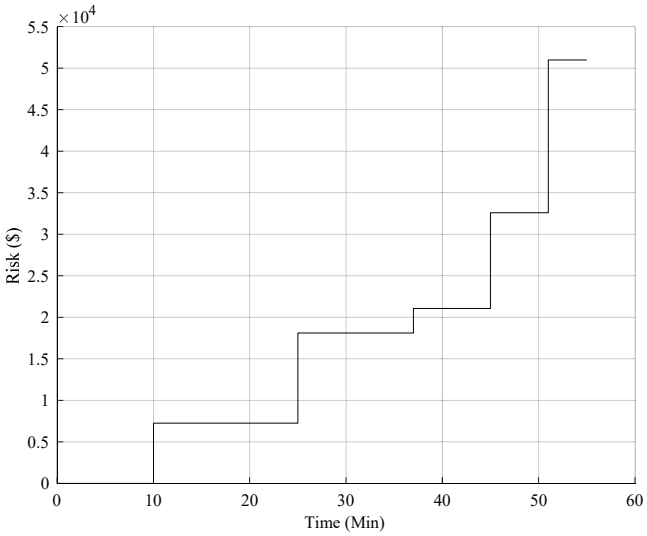


Fig. 5. Overall risk of the SCADA network.

2. Gained the administrator privilege of PC in the 25th minute.
3. Compromised ES by exploiting Vulnerability V_4 in the 37th minute.
4. Succeeded to attack PLC2 via Vulnerability V_8 in the 45th minute.
5. Damaged the blender by tampering the control commands of MO and SW form PLC2 in the 51st minute.

In this experiment we assume that the BN parameters have been learned from historical data. According to Fig. 5 we know that the estimated overall system risk increases as the attack goes deeper in the SCADA network. In other words, the proposed method can reflect the system risk in real time.

Second, we use the proposed machine learning algorithms to improve the risk assessment accuracy. In this experiment, we first learn the BN parameters offline on a small dataset with 50 samples, and then incrementally update the CPT parameters with online observations. These observations are captured when the system is under the same attack scenario as described in the first experiment, and the update frequency is once in every minute. Fig. 6 presents the different risk assessment results when the BN model is trained with different number of online records. It can be observed from Fig. 6 that the risk assessment result is approximating the actual risk level as the number of online observations increases gradually. Hence, this experiment shows that the proposed machine learning algorithms can significantly improve the risk assessment accuracy.

Third, we try to assess the risk caused by unknown attacks. Here we consider the same attack scenario in the first experiment except that V_4 is an unknown vulnerability. The risk assessment result is depicted in Fig. 7. As shown in Fig. 7, when an attacker has exploited an unknown vulnerability to attack the system, the risk value calculated using traditional

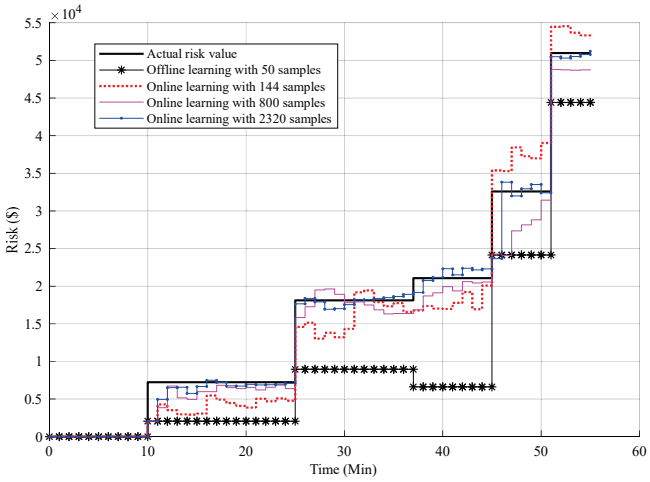


Fig. 6. Risk assessment performance with machine learning capability.

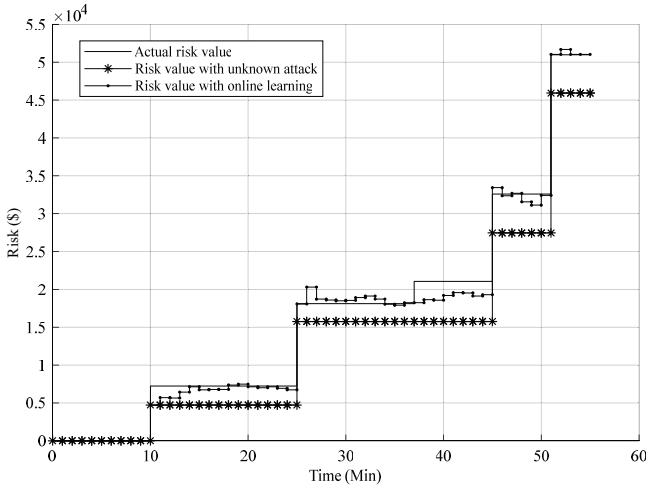


Fig. 7. Risk assessment result for unknown attacks.

BN-based CSRA methods might have large errors; but when we use the proposed BN model which is extended with LNOR gate to assess the system risk, the assessment error tends to be negligible as more and more online observations are available. Therefore, it is demonstrated that our approach is able to assess the risk caused by unknown attacks.

D. Discussion

The experiments conducted in this section show that our proposed method is effective for CSRA in SCADA networks. Actually, this method can also be employed in other circumstances, such as wireless sensor networks and optical communication networks [13], [14]. For example, in optical communication networks, the complex network topologies (e.g., star, line and bus) make it difficult to assess system security status. However, these dependency relationships can be easily depicted using BNs. Then the real-time security risk can be evaluated with the proposed method. In summary, our method is suitable for CSRA in complex networks.

VI. Conclusion

In this paper a Bayesian network-based data-driven cybersecurity risk assessment method is proposed for SCADA networks. The new approach is designed to learn the parameters of the risk assessment model from historical data and online observations, which is more accurate and objective than directly specifying the parameters with expert knowledge. The effectiveness of the proposed approach is validated by a case study on a chemical process SCADA network.

In the future work we plan to add attack nodes into the BN-based CSRA model for more comprehensive security risk assessment.

Acknowledgment

The authors would like to thank the reviewers for their valuable comments. The authors are with the Key Laboratory of Ministry of Education for Image Processing and Intelligent Control, School of Automation, Huazhong University of Science & Technology, Wuhan, Hubei 430074, China. This research was supported by the National Natural Science Foundation of China (No. 61433006 and 61272204).

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